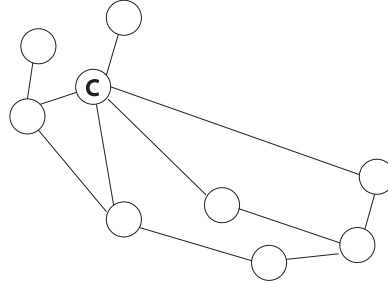


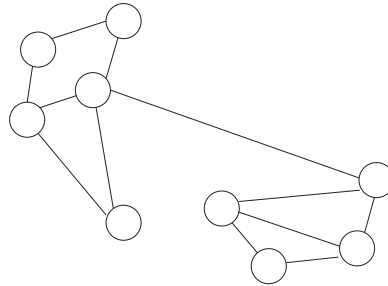
Social Network Analysis

Exam

1. What is the degree of node C in the network below?



2. Does the network below contain any brokers or bridges? If so, shade these nodes/edges. If not, simply write "no."



3. Suppose you're using social network data to develop a marketing campaign. Naturally, you might be interested in targeting the most influential or connected people in that network with advertising or promotional goods. What type of network statistics would give you this sort of information?
4. Give an example of a network clustering algorithm that automatically determines the number of clusters.
5. (TRUE/FALSE) Mark each of the following five statements as being True or False
 - a. A Power-law graph (i.e. scale-free network) is difficult to break apart with random attacks, but is vulnerable to targeted intrusion.
 - b. A star graph is an example of a clique. (A star graph looks like this: )
 - c. The clustering coefficient is a measure of transitivity.
 - d. Most network datasets are examples of Simple Random Sampling.
 - e. Community detection algorithms cannot be used for traditional (non-network) data and traditional data clustering algorithms (like k-means) cannot be used for network data.

6. For the following hypotheses, indicate whether they are at the individual level, dyad level, or network level.
- a. The amount of trade between two countries is correlated with the number of times the leaders of those countries are seen together in photographs.
 - b. Density of a nation's transportation network is related to their Gross Domestic Product.
 - c. People who have more friends on facebook are more successful in creating sales through referrals than those with fewer friends.
 - d. People are more likely to trust others when they've completed a team building seminar together.
7. Why can't we test network hypotheses the same way as our traditional hypotheses, why do we need to use permutation testing?
8. If we use the QAP approach to simulate a p-value for a dyadic level hypothesis, in essence what are we changing about the network in each iteration of that simulation? In other words, if I have a network and I perform a QAP permutation, what have I essentially done?